Nicolas Zilberstein

Pasadena, CA – nmz2@caltech.edu

https://nzilberstein.github.io – https://github.com/nzilberstein

CURRENT POSITION

California Institute of Technology (Caltech) — Pasadena, CA
Postdoctoral Researcher, Computing and Mathematical Sciences — 2026 - Present
PI: Prof. Katie Bouman

EDUCATION

Rice University — Houston, TX
Ph.D. in Electrical and Computer Engineering — 2021 - 2026
Thesis: From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems
Advisor: Prof. Santiago Segarra
Keywords: Inverse problems, Diffusion models, Sampling, Computational imaging

Rice University — Houston, TX
Master of Science in Electrical and Computer Engineering — 2021 - 2023
Thesis: Annealed Langevin dynamics for massive MIMO communications

University of Buenos Aires — Buenos Aires, Argentina
B.S. in Electrical Engineering (with honors) — 2013 - 2020
Thesis: Development of inference algorithms for UWB-GPR signals

WORK EXPERIENCE

Google — Mountain View, CA
Student Researcher at Computational Imaging Team (LUMA) — May 2026 - August 2026
PIs: Kuldeep Purohit, Mojtaba Sahraee-Ardakan and Mauricio Delbracio
Keywords: Diffusion models, Inference-time scaling, Inverse problems.

Center for Computational Neuroscience, Flatiron Institute — New York City, NY
Summer research intern at the Computational Vision group — May 2025 - August 2025
PIs: Eero Simoncelli and Florentin Guth
Keywords: Energy-based model, inverse problems, anisotropic diffusion models.

InterDigital — Los Altos, CA
Research intern - AI for wireless — May 2024 - August 2024
PIs: Akshay Malhotra and Shahab Hamidi-Rad
Keywords: Topological autoencoder, zero-shot stitching, latent space communication.

ExxonMobil — Buenos Aires, Argentina
Data Engineer — 2019 - 2021
Keywords: Front and Back-end, Cloud computing, Agile methodologies.

School of Engineering, University of Buenos Aires — Buenos Aires, Argentina
Research Assistant — 2018 - 2019
PIs: Prof. Bruno Cernuschi-Frias, Prof. Juan Augusto Maya and Prof. Andres Altieri
Keywords: Particle filter, ground penetrating radar imaging, Green's function, sparse Bayesian learning.

AWARDS

Grants

NVIDIA Academic Grant — 2026

AI generated visuals: a solution for image demand in social science research — Rice University
Center for Computational Insights on Inequality and Society (5000 USD) — 2025

Fellowships

Ken Kennedy – HPE Cray Graduate Fellowship — Rice University
One-year graduate fellowships for outstanding graduate students at Rice University — 2024

UBACyT Scholarship (declined) — University of Buenos Aires
Scholarship for undergraduate students to develop a research project — 2018

Honors

Best student paper contest — Asilomar
DAG-PnP [C1] selected as one of the top 8 papers — 2026

Top reviewer — NeurIPS
Selected as a top 8% reviewer based on the area chairs rating — 2026

Top reviewer (Silver) — ICML
Selected as a top reviewer based on the area chairs rating — 2026

Google PhD fellowship nominee — Rice University
One of two nominees selected by the ECE department — 2025

Excellence in Research — Rice University
Award from the ECE section of the Gulf Coast Undergraduate Research Symposium (GCURS) — 2020

Travel grants

ICML Travel Grant — 2026
ICLR, AISTATS Travel Grant — 2025
REA (Rice Engineering Alumni) Graduate Student Travel Grant — 2025

PUBLICATIONS

Journals

[J1] Haoming Chen, Nicolas Zilberstein, and Santiago Segarra. Prior-informed flow matching for graph reconstruction. *Trans. Mach. Learn. Res. (TMLR)*, 2026

[J2] Nicolas Zilberstein, Ashutosh Sabharwal, and Santiago Segarra. Solving linear inverse problems using higher-order annealed Langevin diffusion. *IEEE Trans. Signal Process.*, 2024

[J3] Fernando Gama, Nicolas Zilberstein, Martin Sevilla, Richard G Baraniuk, and Santiago Segarra. Unsupervised learning of sampling distributions for particle filters. *IEEE Trans. Signal Process.*, 2023

[J4] Nicolas Zilberstein, Chris Dick, Rahman Doost-Mohammady, Ashutosh Sabharwal, and Santiago Segarra. Annealed Langevin dynamics for massive MIMO detection. *IEEE Trans. Wireless Commun.*, 2022

[J5] Nicolas Zilberstein, Juan Augusto Maya, and Andres Altieri. A BCS microwave imaging algorithm for object detection and shape reconstruction tested with experimental data. *Electr. Lett.*, 57(2):88–91, 2021

Conference Papers

[C1] Nicolas Zilberstein, Ali Azizpour, Gonzalo Mateos, and Santiago Segarra. DAG-PnP: Plug-and-play causal discovery with diffusion priors. In *Asilomar Conf. Signals, Syst. and Comp.* IEEE, 2026. **Best student paper contest (top 8)**

[C2] Nicolas Zilberstein, Morteza Mardani, and Santiago Segarra. Flow map denoisers: Traversing the distortion-perception plane for inverse problems. *Advances in Neural Inf. Process. Syst. (NeurIPS)*, 2026

[C3] Nicolas Zilberstein, Santiago Segarra, Eero Simoncelli, and Florentin Guth. Normalized energy models for linear inverse problems. *Intl. Conf. on Machine Learning (ICML)*, 2026

[C4] Nicolas Zilberstein, Santiago Segarra, and Luiz Chamon. Sampling with shielded Langevin Monte Carlo using navigation potentials. In *Asilomar Conf. Signals, Syst. and Comp.* IEEE, 2025

[C5] Nicolas Zilberstein, Morteza Mardani, and Santiago Segarra. Repulsive latent score distillation for solving inverse problems. *Intl. Conf. Learn. Repr. (ICLR)*, 2025

[C6] Ali Azizpour, Nicolas Zilberstein, and Santiago Segarra. Scalable implicit graphon learning. *Intl. Conf. on Artif. Intell. and Stat. (AISTATS)*, 2025

[C7] Benjamin Cox, Sara Perez-Vieites, Nicolas Zilberstein, Martin Sevilla, Santiago Segarra, and Victor Elvira. End-to-end learning of Gaussian mixture proposals using differentiable particle filters and neural networks. *IEEE Intl. Conf. Acoust., Speech and Signal Process. (ICASSP)*, 2024

[C8] Nicolas Zilberstein, Ananthram Swami, and Santiago Segarra. Joint channel estimation and data detection in massive MIMO systems based on diffusion models. *IEEE Intl. Conf. Acoust., Speech and Signal Process. (ICASSP)*, 2024

[C9] Martin Sevilla*, Nicolas Zilberstein*, Benjamin Cox, Sara Perez-Vieites, Victor Elvira, and Santiago Segarra. State and dynamics estimation with the Kalman–Langevin filter. In *Asilomar Conf. Signals, Syst. and Comp.*, pages 53–57. IEEE, 2023

[C10] Nicolas Zilberstein, Chris Dick, Rahman Doost-Mohammady, Ashutosh Sabharwal, and Santiago Segarra. Accelerated massive MIMO detector based on annealed underdamped Langevin dynamics. In *Graph Signal Processing Workshop*, 2023

[C11] Nicolas Zilberstein, Chris Dick, Rahman Doost-Mohammady, Ashutosh Sabharwal, and Santiago Segarra. Accelerated massive MIMO detector based on annealed underdamped Langevin dynamics. In *IEEE Intl. Conf. Acoust., Speech and Signal Process. (ICASSP)*, 2023

[C12] Nicolas Zilberstein, Chris Dick, Rahman Doost-Mohammady, Ashutosh Sabharwal, and Santiago Segarra. Detection by sampling: Massive MIMO detector based on Langevin dynamics. In *European Signal Process. Conf. (EUSIPCO)*, pages 1651–1655, 2022

[C13] Nicolas Zilberstein, Chris Dick, Rahman Doost-Mohammady, Ashutosh Sabharwal, and Santiago Segarra. Robust MIMO detection using hypernetworks with learned regularizers. In *European Signal Process. Conf. (EUSIPCO)*, pages 1626–1630. IEEE, 2022

[C14] Fernando Gama, Nicolas Zilberstein, Richard G Baraniuk, and Santiago Segarra. Unrolling particles: Unsupervised learning of sampling distributions. In *IEEE Intl. Conf. Acoust., Speech and Signal Process. (ICASSP)*, pages 5498–5502. IEEE, 2022

[C15] Nicolas Zilberstein and Bruno Cernuschi-Frias. Particle filter with unknown noise statistics and with prior knowledge. In *2018 Argentine Conference on Automatic Control (AADECA)*, pages 1–6. IEEE, 2018

Workshop Papers

[W1] Victor M. Tenorio, Nicolas Zilberstein, Santiago Segarra, and Antonio G. Marques. Graph guided diffusion: Unified guidance for conditional graph generation. *NeurIPS workshop NPGML*, 2025

[W2] Nicolas Zilberstein, Akshay Malhotra, Yugeswar Deenoo, and Shahab Hamidi-Rad. Topology preserving regularization for independent training of inter-operable models. *NeurIPS workshop UniReps*, 2024

Patents

1. "Systems and methods for statistics based and topology based interoperable AI/ML model monitoring" 2024
Akshay Malhotra, Yugeswar Deenoo, Nicolas Zilberstein, Shahab Hamidi-Rad, Mohamed Salah Ibrahim
InterDigital, US Patent Ref. 18889096

Preprints

[P1] Ali Azizpour, Nicolas Zilberstein, and Santiago Segarra. Model-driven graph contrastive learning. *arXiv preprint arXiv:2506.06212*, 2025

*Equal contribution

TEACHING EXPERIENCE

Department of Electrical and Computer Engineering, Rice University Houston, TX
Teaching Assistant

- ELEC 573 Network Science and Analytics, *Graduate course* Fall 2022, 2024
- ELEC 301 Signals and Systems, *Undergraduate course* Spring 2022, 2023, 2024, 2025

Graduate school of Computer Engineering, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- Deep Learning Fall 2021

Department of Physics, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- 82.02: Physics II (Electromagnetism) Spring 2020, Fall 2020

Department of Electronic Engineering, University of Buenos Aires Buenos Aires, Argentina
Teaching Assistant

- 86.55: Detection and Estimation theory Spring 2018, 2019

PROFESSIONAL ACTIVITIES

Invited talks

From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems Dec 2025
Universidad de la Republica Uruguay

From Guided Sampling to Learned Solvers: Diffusion Models for Inverse Problems Nov 2025
Yusu Wang's group at *UCSD*

Towards fast samplers for solving inverse problems using diffusion models June 2024
Pendulum Systems

Organization

Special Session: Generative AI for Graphs Asilomar 2026
Organizer and Technical Chair

Workshops

Advances of Approximate Bayesian Inference (AABI) 2025 Singapore
Repulsive Latent Score Distillation for Solving Inverse Problems [C5] presented in the fast track

Khipu AI 2025 Santiago, Chile
The largest meet-up gathering students, researchers, and innovators in the field of Artificial Intelligence and Machine Learning across Latin America

LOGML Summer School 2024 London, UK
Workshop that aims to bring together mathematicians and computer scientists to work together on a variety of problems at the intersection of geometry and machine learning

Graph Signal Processing Workshop 2023 Oxford, UK
Workshop on graph signal processing. Poster presentation.

Reviewer

Journals: IEEE Comm. Letter; IEEE Trans. Cogn. Commun.; IEEE Trans. Signal Process.; IEEE Trans. Commun.
Conferences: ICASSP, EUSIPCO, ICC, SAM, CAMSAP, NeurIPS (2024, 2025, 2026), ICLR (2026), ICML (2026), AAAI (2026)

Memberships

IEEE Member 2021 - Present
IEEE Signal Processing Society Member 2021 - Present

STUDENT ADVISEMENT

Graduate

Ali Azizpour

Undergraduate

Haoming Chen	Summer 2025
Uros Males, Mila Bokan	Summer 2023

OTHER SKILLS

Languages: Spanish (native), English (fluent)

COMMUNITY SERVICES

Evaluator at Gulf Coast Undergraduate Research Symposium (GCURS)	2024, 2025
Mentor of incoming ECE PhD students - Data science track	2024
Organizer, LatAm RICE Networks, Robots, and ML Minisymposium, Rice Univ.	2021
Member of the Council of the Faculty of Engineering, Univ. of Buenos Aires	2019 - 2021
Member, Advisory Board, Dept. of Electronics, Univ. of Buenos Aires	2018 - 2021
Member, Curricular Committee, Electronic Engineering Degree, Univ. of Buenos Aires	2017 - 2021